HANSEL_MESH Code And Network Explainer

€ • ‚ ƒ „ …†

‡ˆ‰: 2026-06-03

1. Š‹Œ•Ž•

HANSEL_MESH• ‘’“”•–—˜•™š› ‹œ•žŸ ¡¢£”˜¤¥¦Š‹Œ•™. §¨© ª•«¬ basePi-š®LAN¯‹°±²³, base/head/node1/node2• BATMAN-adv¤´ mesh•µ¶¦™.

·¸¹º:

```
§¨©ª•«
|
basePi
|
BATMANMesh
|
node2
|
node1
|
headPi + Camera + Motor
```

»Ž¦ ,¼•½:

- ¾ base• ƒ¿À„ÁÂÃ˜ÄÅ•™.
- ¾ node1/node2•ÆÇ¡ÈÉÊ relayË ÌÍÎ BATMAN-adv Layer 2 relay‹ Ï‡¦™.
- ¾ ¨Ð ƒ¿ÑÒÓÎ ÔÕ¬ bat0 ¥•† end-to-end‹ ¹Ö™.
- ¾ “× ÏØÙ• §ˆ•†• ¨Ú©ÛÜÝ AP ¤Þ¬ §“˜ÄÅ•™.
- ¾ headß¨à¦™. node1/node2• headË ßáâ‹¢ ã–/äåæÐ¦™.

2. çÜ §¨

```
configs/
base.env
head.env
node1.env
node2.env

scripts/
start_mesh.sh
stop_mesh.sh
check_mesh.sh
setup_base_gateway.sh
setup_laptop_mesh_routes.sh
setup_mesh_route_to_laptop.sh
start_camera_stream.sh
receive_camera_stream.sh

controller/
mesh_control_client.py

robot/
mesh_control_server.py
motor_driver.py
__init__.py

docs/
tomorrow_drive_camera_runbook.md
motor_control_quickstart.md
camera_control_quickstart.md
next_field_test_checklist.md
```

3. žŸèéŽ

3.1 IPêë

”ì	íîïÉð	IP
laptop	LAN adapter	192.168.60.2/24
base	eth0	192.168.60.1/24
base	bat0	192.168.50.1/24
head	bat0	192.168.50.10/24
node1	bat0	192.168.50.11/24
node2	bat0	192.168.50.12/24

192.168.60.0/24• ª•«Ñ base ñÉ• ò¡óÉ™. 192.168.50.0/24• BATMAN mesh ¥•† ñ„˜• ‹œóÉ™.

3.2 BATMAN-adv •¡

BATMAN-adv• Linux kernel module‹ Ï‡˜• Layer 2 mesh ôõÉ™. Linux kernel documentation• ö÷ø batman-adv• IP routing tableÁ ãù™ú• ôõÉ ÌÍÎ Ethernet frameÁ Layer 2•† ûü˜ý, µ¶ ªÜþÉ˜ÿ• ËÕ switch É ¦™. öÎ† IPv4, IPv6, DHCP ¬Õ¥ Š‹ É mesh topology • ÔàÁ ÄÅ•™.

ÉŠ‹Œ••† »Ž¦• :

- ¾ wlan0• IBSS/ad-hoc ® ë ß¦™.
- ¾ wlan0•• IP¢ ÄÅ•™.
- ¾ bat0ËÕ¥ÆÇ¡ÈÉÊÉ • žŸ íîïÉð™.
- ¾ IP• bat0•ß è¶¦™.
- ¾ nodeË» • ì²ø BATMAN-advË next-hopÁ©Ï ® ¦™.

3.3 batctl £í ƒ¿

batctl¬ batman-adv kernel moduleÕ ¢£í˜• §™. batctl documentation¬ neighbor tableÑ originator tableÁ žÂ ãù É Ñ· Ä next-hopÁ ™³, ƒ¦™.

ƒ¿ :

```
sudo batctl n
sudo batctl o
```

ÂÃ :

- ¾ batctl n: “× ãù ŸËÞ¦ neighbor
- ¾ batctl o: mesh û originator-· Ä selected next-hop
- ¾ batctl o•† *Ë!¬"¬ BATMANÉ® ¦“× best next hopÉ™.
- ¾ · ÄË headí# selected Nexthop É node1/node2 MAC Éø multi-hop relay »É™.

3.4 start_mesh.sh

ë :

1. Wi-Fi $%Â
2. batman-adv kernel module load
3. hostapd/dnsmasq/wpa_supplicant &'(ËÞ†)ð*¡

4. ¤+bat0 ,

5. wlan0¢meshpoint -• IBSS‹ §ˆ

6. bat0 .ˆ

7. wlan0¢ BATMAN hard interface‹ æË

8. bat0• ³* IPè¶

Pi 3À”Wi-Fi• /ð••†meshpointË ÌÍÎ IBSS¢ñ„0™. 12† start_mesh.sh• MESH_MODE=auto•† ËÞ¦ 3Ü¢ £í˜³ IBSS fallbackÁ 4¦™.

»Ž :

```
MESH_IF=wlan0
BAT_IF=bat0
MESH_ID=HANSEL_MESH
MESH_FREQ=2437
IBSS_BSSID=02:12:34:56:78:9a
IP_ADDR=192.168.50.x
NETMASK_CIDR=24
```

3.5 base gateway

setup_base_gateway.sh• base• eth0¢ ª•«ò¡ó¯‹ ,*¦™.

```
base eth0 = 192.168.60.1/24
base bat0 = 192.168.50.1/24
```

-¦ Linux IPv4 forwardingÁ 5ˆ ¦™.

```
sysctl -w net.ipv4.ip_forward=1
```

base•¶¤† ÆÇ¡ÈÉÊ relayË ÌÍ™. %67 kernel routing¯‹ eth0-bat0 ñÉ¢ °±¦™.

3.6 laptop route

setup_laptop_mesh_routes.sh• ª•« LAN adapter¢ 192.168.60.2/24‹ ,*˜³, mesh êëÁ base‹ 8™.

```
ip route replace 192.168.50.0/24 via 192.168.60.1
```

É routeË 9: ª•«É 192.168.50.10 head‹ ãù UDP/pingÁ ; ™.

3.7 head route back to laptop

head•† setup_mesh_route_to_laptop.sh¢ 4˜ø 192.168.60.0/24‹ Ë• routeË base bat0¢ à¦™.

```
ip route replace 192.168.60.0/24 via 192.168.50.1 dev bat0
```

ÒÓÎ ÔÕ¬ head•† ª•« 192.168.60.2‹ UDP <Ÿ˜=‹ É reverse routeË >Ž˜™.

4. #Éî? -žŸ?

4.1 ¨Ð ƒ¿ UDP JSON

ª•«• controller/mesh_control_client.py• UDP JSON packetÁ 8™.

@:

```
{
"seq": 12,
"target": "head",
"command": "forward",
"source": "operator",
"time": 1780400166.3324077,
"speed": 0.4
}
```

>Ü:

>Ü	•
seq	<Ÿ 6A
target	head, node1, node2
command	forward, stop &ƒ¿
source	operator
time	<Ÿ BC
speed	® >Ü, 0.0~1.0å scale

UDP port:

```
7000
```

UDP¢ D• Éš:

¾ ¨Ð ƒ¿¬ EŸ Õ Ë »Ž˜™.

¾ F2G ƒ¿Á TCP ×û< •H ™watchdog¯‹ Iæ• JÉ Kû˜™.

¾ packet lossË 9 livemodeË ¤ ¯‹ ƒ¿Á´L <Ÿ¦™.

4.2 ¨Ð ƒ¿ ¹º

```
controller/mesh_control_client.py
-> UDP JSON
-> laptop LAN 192.168.60.2
-> base eth0 192.168.60.1
-> base bat0 192.168.50.1
-> BATMAN selected next-hop
-> target bat0
-> robot/mesh_control_server.py
-> robot/motor_driver.py
-> GPIO motor driver
```

4.3 livemode ÎMN

--target all‰O ƒ¿P :

Q	head	node1/node2
w	forward	forward
s	backward	backward
a	left	stop
d	right	stop
q	forward_left	slow_forward
e	forward_right	slow_forward
z	backward_left	slow_backward
c	backward_right	slow_backward
x/space	stop	stop

R¨à¬ headß 4˜³, node• headË ßá â‹¢ öÎË¤ ¥Â ã–/ äå/*Äß ¦™.

node†S•† É»Kû”ì¢T™. nodeË ‹ left, right, forward_left ¬¨àƒ¿Á ãù Ì stop, slow_forward, slow_backward‹ U¦™.

4.4 ÒÓÎ ÔÕ UDP H.264

head•†:

```
rpicam-vid -t 0 --nopreview --width 640 --height 480 --framerate 15
--codec h264 --inline
--bitrate 1200000 -o udp://192.168.60.2:5600
```

ª•«•†:

```
./scripts/receive_camera_stream.sh 5600
```

VW:

¾ H.264 elementary stream

¾ UDP port 5600

¾ --inline¯‹ SPS/PPS¢ ¤ ¯‹ XYÂ decoderË» ŸÂ L§˜¤Z Y

¾ mesh Uâ•†•ÂÕ -bitrate¢ [\K*ˆÁM®¦™.

æ] B‡^:

```
320x240, 10fps, 600kbps
```

Ë_` ¡K*^:

```
640x480, 15fps, 1200kbps
```

5. §Ïè éŽ

5.1 a‰ë

robot/mesh_control_server.py:

¾ UDP socket bind

¾ JSON packet parsing

¾ watchdog timeout

¾ dry-run Ä•

¾ motor_driver• command ûü

robot/motor_driver.py:

¾ GPIO pin map

¾ PWM motor output

¾ encoder polling

¾ PID control loop

¾ head servo

¾ detach servo

¾ head front DC motors

¾ node command safety normalization

controller/mesh_control_client.py:

¾ ª•« ¨Ð—b

¾ line mode

¾ live mode
¾ target UDP JSON <Ÿ
¾ --target all •† head/node ƒ¿ P

5.2 GPIO pin map

cž drive motor:

¤Þ	GPIO
ENA left PWM	18
IN1 left	23
IN2 left	24
ENB right PWM	13
IN3 right	27
IN4 right	22

encoder:

¤Þ	GPIO
LEFT_ENC_A	20
LEFT_ENC_B	21
RIGHT_ENC_A	16
RIGHT_ENC_B	26

servo:

¤Þ	GPIO
detach servo	6
head servo	17

head front DC motors:

¤Þ	GPIO
FRONT_ENA	12
FRONT_IN1	4
FRONT_IN2	25
FRONT_ENB	19
FRONT_IN3	5
FRONT_IN4	7

5.3 §Ï Ódĺe

head:

¾ fMghi RPM $É‹ ¨à¦™.
¾ jk ³é èP DC3î 2é ¤l ¯‹ 4hi- ¬ôà/PWM ratio¢ öÎ ™.
¾ forward_left ømk target CPSË [³ FÖk target CPSË n™.
¾ left ømk¬ o–, FÖk¬ û–Â ©¡ ¨à¦™.

node1/node2:

¾ ã–/o–/*Äß 4¦™.
¾ ¨à ƒ¿Á ¯ø †SÀè•† Kû˜ U¦™.

¾head¨à»•• slow_forward -• slow_backward‹ öÎ ™.

5.4 pçq ¡

encoder• A/B gŸr• quadrature transitionÁs•™.

Õ :

```
state = (A << 1) | B
```

transition:

```
transition = (last_state << 2) | new_state
```

forward count tË transition:

```
0001, 0111, 1110, 1000
```

backward count uv transition:

```
0010, 1011, 1101, 0100
```

RPi.GPIO event detect• OS/kernel ¨w•†… Ëx 9 polling thread¢ ñ„¦™.

¤l polling interval:

```
0.001s
```

5.5 PID 9‹ã

9 ¤:

```
CONTROL_INTERVAL = 0.05s
```

C ¤•†:

```
delta_count = current_count - previous_count
measured_cps = abs(delta_count) / dt
error = target_cps - measured_cps
integral += error * dt
derivative = (error - previous_error) / dt
```

feed-forward:

```
base_pwm = MIN_PWM + (target_cps / max_cps) * (MAX_PWM - MIN_PWM)
```

PID output:

```
pid_output = Kp * error + Ki * integral + Kd * derivative
pwm_request = clamp(base_pwm + pid_output, 0, 100)
```

PWM ramp ¦:

```
max_delta = PWM_RAMP_PER_SEC * dt
pwm = clamp(pwm_request, previous_pwm - max_delta, previous_pwm + max_delta)
```

É§¨• Éš:

¾ feed-forward• ·¸å •y• ¤l PWMÁ z÷ c¦™.

¾ PID• è˜/{|/ î¡û} $É¢ *¦™.

¾ ramp ¦¬~•¦ û€/ ¢"í™.

¾ integral limit¬ wind-upÁ ôÄ¦™.

¤l^:

aÎ î	^
FULL_SPEED_CPS	800
MIN_PWM	25
MAX_PWM	100

aÎ î	^
KP	0.035
KI	0.015
KD	0
INTEGRAL_LIMIT	500
PWM_RAMP_PER_SEC	220

5.6 Kû ¤Þ

¾ UDP command watchdog: ¤ I 0.5• ÏK ƒ¿É Ù¯ø stop
¾ Ctrl+C Ð, B stop û<
¾ GPIO cleanup
¾ encoder polling error ƒ. B motor stop B
¾ node ¨à ƒ¿ B/ U
¾ --dry-run¯‹ GPIO ÙÉ „•… ß/ð• ËÞ
¾ duplicate GPIO pin check

6. ÎÉ†‡¡

Python ¸ˆ ÎÉ†‡¡:

¾ socket: UDP < Ÿ
¾ json: command packet encoding/decoding
¾ time: timestamp, control loop timing
¾ threading: encoder polling loop - PID control loop
¾ argparse: CLI option parsing
¾ select, termios, tty: live keyboard input
¾ dataclasses: pin/config §¨

Raspberry Pi ÎÉ†‡¡:

¾ RPi.GPIO: motor driver direction pin, PWM, encoder input
¾ pigpio: head servo pulse width control

‰è BðŠ §:

¾ batctl: BATMAN-adv Õ £í
¾ iw, iwconfig: IBSS/ad-hoc ®,* £í
¾ ip: interface/IP/route ,*
¾ rpicam-vid: Camera Module 3 ÔÕ <Ÿ
¾ ffplay, gst-launch-1.0, vlc: ª•« ÔÕ Ÿ

7. Ž 4¹º

7.1 mesh B‡

```
start_mesh.sh
-> config source
-> Wi-Fi manager stop
-> wlan0 IBSS/ad-hoc join
```

```
-> bat0 create
-> batctl if add wlan0
-> bat0 IP assign
```

7.2 ¨Ð †SB‡

```
mesh_control_server.py
-> build_robot_controller(role)
-> GPIO setup or dry-run
-> UDP bind 0.0.0.0:7000
-> receive packet
-> parse JSON
-> watchdog update
-> motor_driver.handle_command()
```

7.3 motor command ¡

```
handle_command()
-> normalize command
-> if node: steering command -> straight/stop
-> command_map lookup
-> set target CPS and direction
-> PID loop changes PWM
```

7.4 camera ¡

```
receive_camera_stream.sh on laptop
-> ffplay/gstreamer/vlc wait UDP 5600

start_camera_stream.sh on head
-> rpicam-vid
-> H.264 UDP
-> 192.168.60.2:5600
```

8. BATMAN relay ‹% ôŒ

Ë_` ¡:

```
base -> head direct
```

ÉO batctl o• selected nexthop¬ · ÄMACÑ Á ™.

•¡ ì:

```
base -> node2 -> node1 -> head
```

ÉO base• batctl o•† head originator• selected nexthopÉ node2 -• node1 kMAC¯‹ hŽø relay™.

•t ƒ¿:

```
ping 192.168.50.10
sudo batctl n
sudo batctl o
```

9. … Â±

9.1 base- ª•« °±É•'

base:

```
sudo ./scripts/setup_base_gateway.sh
```

ª•«:

```
sudo ./scripts/setup_laptop_mesh_routes.sh enx00e04c68070e
```

£í:

```
ping -c 4 192.168.60.1
ping -c 4 192.168.50.10
```

9.2 ÒÓÎ busy

```
sudo pkill -f rpicam-vid
sudo pkill -f libcamera-vid
sudo pkill -f rpicam-hello
sudo pkill -f libcamera-hello
sudo pkill -f rpicam-still
sudo pkill -f libcamera-still
```

9.3 motor server B‡ ’

£í:

¾ sudo‹ 40•Ë

¾ RPi.GPIO ,ì¶è

¾ encoder “ ®

¾ Éû 9†SË GPIO¢ ”³ •Ä

¾ pin conflict ÓBÄ

9.4 ôà ´ê

Uâ ‹ *:

```
HANSEL_LEFT_REVERSE=yes sudo -E python3
~/HANSEL_MESH/robot/mesh_control_server.py --role
head
HANSEL_RIGHT_REVERSE=yes sudo -E python3
~/HANSEL_MESH/robot/mesh_control_server.py --role
head
```

9.5 ÔÕ •–

[¬ ,*¯‹ B‡:

```
WIDTH=320 HEIGHT=240 FPS=10 BITRATE=600000
~/HANSEL_MESH/scripts/start_camera_stream.sh
192.168.60.2 5600
```

10. µ³ ©,

¾ Linux kernel batman-adv documentation: https://www.kernel.org/doc/html/v4.17/networking/batman-adv.html

¾ batctl upstream repository and man page text: https://github.com/open-mesh-mirror/batctl

¾ batctl HTML man page: https://downloads.open-mesh.org/batman/manpages/batctl.8.html

¾ Previous HANSEL_GRETEL repository referenced for motor pin map and control structure: https://github.com/SSING-rloz/2026-_HANSEL_GRETEL